

Ruei-Hong (Neil) Ni

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EDUCATION

National Taipei University
Bachelor of Communication Engineering

New Taipei, Taiwan
Sep. 2025 – Present

Mingdao High School
Gifted and Talented Program

Taichung, Taiwan
Aug. 2019 – Jun. 2025

EXPERIENCE

High School Robotics Team Coaching
FRC Team 7130, Mingdao High School

Feb. 2025 – Present
Taichung, Taiwan

- Spearheaded the deployment of a practice field Field Management System (adapted from Team 254), creating a competition-grade simulation environment.
- Configured enterprise-grade network infrastructure for the arena, managing Cisco switches, VLANs, and DHCP to ensure reliable, zero-latency device connectivity.
- Designed and implemented a comprehensive technical training curriculum, ensuring members mastered engineering fundamentals prior to the competition season.
- Mentored student drive teams on tactical robot operations and strategic decision-making to maximize competitive performance.

PROJECTS

HAPPINESS SEEKER (Interactive Line Bot) | API Integration, System Debugging

Nov. 2025

- Integrated the Line Messaging API to handle concurrent user requests and build an interactive conversational interface.
- Orchestrated the connection with the Gemini LLM API for dynamic text generation, utilizing AI-assisted coding to accelerate development.
- Successfully troubleshooted and resolved complex backend issues, including server response errors and API versioning conflicts during deployment.

FIRST Robotics Competition (Team 7130) | Java, Embedded Programming

June 2022 – June 2023

- Designed a modular Finite-State Machine (FSM) to streamline robot actions, improving code maintainability and debugging efficiency.
- Developed scalable Java code for the NI RoboRIO embedded system, coordinating motor output with sensor data for a 120lb competitive robot.
- Optimized robot motion planning using PID controllers and odometry, significantly reducing path tracking errors during autonomous periods.
- Leveraged OpenCV to engineer a vision-based localization system, allowing the robot to autonomously identify and align with field targets.

TECHNICAL SKILLS

Proficient: Java

Familiar: C/C++, JavaScript, HTML/CSS

Frameworks & APIs: Line Messaging API, Gemini API, OpenCV

Networking & Infrastructure: Cisco Configuration, VLAN, DHCP, WebSockets

Embedded Systems: NI RoboRIO, Arduino, Raspberry Pi, ESP32

Developer Tools: Git, Visual Studio Code, IntelliJ IDEA